

The title

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Abstract

15 One or two sentences providing a **basic introduction** to the field, comprehensible to a
16 scientist in any discipline. Two to three sentences of **more detailed background**,
17 comprehensible to scientists in related disciplines. One sentence clearly stating the **general**
18 **problem** being addressed by this particular study. One sentence summarizing the main
19 result (with the words “**here we show**” or their equivalent). Two or three sentences
20 explaining what the **main result** reveals in direct comparison to what was thought to be
21 the case previously, or how the main result adds to previous knowledge. One or two
22 sentences to put the results into a more **general context**. Two or three sentences to provide
23 a **broader perspective**, readily comprehensible to a scientist in any discipline.

24 *Keywords:* keywords

25 Word count: X

The title

Historically, Behavior Research Methods (BRM) has served as a venue for the dissemination of tools, analytic procedures, experimental paradigms, and computational workflows that shape how research is conducted and interpreted. As empirical research has become increasingly computational and data-intensive, the accessibility and interpretability of methodological components have become central to the evaluation of scientific claims making it necessary for methodological infrastructure, which can be examined, implemented, and extended across research domains.

The continued challenges to normalize transparency and open access make it increasingly difficult to assess how findings were produced or to assess whether they can be independently evaluated. In this context, transparency is not only a reporting journal preference, but a condition for interpreting and evaluating research.

In response to these challenges, stakeholders (journals, funders, research institutes) find it necessary to clarify expectations for increased transparency in research reporting, tool development, and experimental procedures. The Transparency and Openness Promotion (TOP) Guidelines provided such a structure which allowed journals to formalize transparency standards. Rather than defining transparency as a single requirement, the guidelines establish levels of implementation that distinguish among disclosure, enforcement, and verification. This structure allowed journals to clearly define expectations which are appropriate for their outlet and accommodate diverse methodological approaches in a consistent and interpretable manner.

These developments are directly relevant to BRM's scope and publication model. The journal publishes methodological contributions intended for implementation and reuse, transparency and accessibility are integral to the interpretability and utility of published work. This tutorial provides an overview of the TOP Guidelines, explains the changes

introduced in the 2025 revision, and describes how BRM’s editorial policies align with the updated framework. The goal is to clarify expectations for authors and reviewers and to support consistent application of transparency standards within the journal.

The 2025 Update

Origins of TOP (2015). Following their introduction in 2015, the TOP Guidelines were adopted by journals, publishers, and research organizations across disciplines. This adoption demonstrated broad support for formalizing transparency standards at the journal level. At the same time, implementation across diverse editorial and methodological contexts revealed areas where the original framework lacked sufficient clarity to support consistent interpretation and application.

Although the original development of the TOP Guidelines was multidisciplinary, the contributing authors were primarily drawn from the social and behavioral sciences. As a result, the framework was initially implemented in research environments with relatively similar methodological assumptions and infrastructure. Over time, adoption expanded to disciplines with substantially different research designs, data structures, and standards for documentation and dissemination. This broader application highlighted ambiguities in how transparency standards should be interpreted and operationalized across contexts.

Editors and publishers reported particular challenges in translating endorsement of the guidelines into consistent editorial policy. While the framework provided a shared vocabulary for transparency, distinctions among levels of implementation were not always sufficiently precise to support uniform interpretation. These challenges reflected the difficulty of defining general standards that could accommodate variation in research practice while remaining operationally meaningful.

In response to these developments, the Center for Open Science coordinated a revision of the guidelines informed by feedback from editors, researchers, and other stakeholders. The

76 resulting update reflects accumulated experience with the framework and aims to clarify how
77 transparency standards can be defined and implemented within contemporary research and
78 publication environments.

79 **What Changed and Why.** The 2025 revision introduces structural changes
80 intended to clarify the relationship between transparency and verification. In the original
81 framework, verification was incorporated as the highest level of implementation within
82 individual transparency standards. Experience with this structure demonstrated that
83 disclosure and verification represent distinct editorial and methodological functions that
84 benefit from explicit differentiation.

85 The revised framework distinguishes between research practices and verification
86 practices. Research practices address the accessibility and documentation of methodological
87 components, including data, code, materials, and analytic procedures. Verification practices
88 address the extent to which research outputs can be independently evaluated using the
89 available documentation and resources. This distinction clarifies that transparency facilitates
90 verification but does not, by itself, ensure that verification has occurred.

91 This restructuring also reflects changes in research infrastructure and editorial
92 workflows. The increasing availability of trusted repositories, executable computational
93 environments, and standardized documentation practices has expanded the feasibility of
94 sharing and evaluating research components. The revised framework aligns transparency
95 standards with these developments while maintaining flexibility across research domains.

96 More broadly, the revision reflects an evolution in how transparency standards are
97 conceptualized and implemented. Rather than representing a departure from the original
98 framework, TOP 2025 refines and extends it in response to experience gained through
99 widespread adoption and application.

100 **Problems and Barriers of the 2015 TOP Guidelines.** Experience with TOP
101 2015 highlighted practical constraints affecting implementation at the journal level.

Although many journals formally endorsed the guidelines, fewer adopted policies requiring enforcement or verification. In many cases, endorsement functioned as a public commitment to transparency without corresponding changes to editorial procedures.

These patterns reflected structural constraints within editorial workflows. Implementing transparency requirements involves not only policy revision but also procedures for evaluating compliance and managing exceptions. Without dedicated infrastructure or clearly defined operational standards, disclosure requirements were more readily implemented than policies requiring verification.

Ambiguities within the original framework further contributed to variability in implementation. Differences in how levels were defined across standards made consistent interpretation difficult, particularly in multidisciplinary journals. Verification, in particular, lacked a uniform operational definition, limiting its feasibility as a broadly adopted editorial practice.

These experiences did not reflect opposition to transparency, but rather the complexity of translating general principles into operational editorial standards. The 2025 revision addresses these limitations by clarifying terminology, separating transparency from verification, and aligning the framework more closely with contemporary research and publication practices. In doing so, the updated guidelines provide a more precise foundation for journals seeking to implement and communicate transparency standards.

BRM's Longstanding Commitment to Transparency

Since its founding, Behavior Research Methods has focused on publishing methodological contributions intended for implementation, evaluation, and reuse. The journal provides a venue for disseminating experimental paradigms, analytic procedures, software tools, and measurement approaches that support empirical research. These contributions function as shared methodological infrastructure, enabling researchers to apply,

extend, and evaluate research procedures across contexts.

As methodological practices have evolved alongside advances in computing and data analysis, the journal has adapted its scope to support emerging forms of research infrastructure. Many articles provide not only conceptual descriptions but also executable implementations and detailed documentation, allowing methodological procedures to be examined and applied directly. This publication model supports the interpretability and evaluability of methodological claims by making the procedures underlying those claims accessible.

These practices developed prior to the formalization of transparency standards under initiatives such as the TOP Guidelines. The journal's emphasis on methodological clarity, documentation, and accessibility reflects longstanding expectations that published methods should be sufficiently specified to permit independent implementation and evaluation. In this way, transparency has functioned as an implicit feature of the journal's publication model.

The adoption of the TOP 2025 framework provides a standardized structure for articulating and formalizing these expectations. By aligning editorial policies with the updated guidelines, BRM clarifies how transparency and verification practices apply within the journal's methodological scope. This alignment supports consistent interpretation of transparency standards while maintaining continuity with the journal's established publication practices.

The following section describes how BRM's editorial policies correspond to the TOP 2025 framework and how these standards are applied in practice.

References

Table 1

TOP 2015 transparency standards

Area	Level 0
Data citation	No mention of data citation.
Data transparency	Data sharing is encouraged, or not ment
Analytical code transparency	Code sharing is encouraged, or not ment
Materials transparency	Materials sharing is encouraged, or not m
Reporting guidelines AKA Design and Analysis Transparency	No mention of reporting guidelines.
Study preregistration	Journal says nothing
Analysis plan preregistration	Journal says nothing
Replication	Journal says nothing
Publication Bias	Journal says nothing
Open science badges	

Table 2
Research Practices

Practice	Level 1: Disclosed
Study Registration	Authors stated whether or not a study was registered—and, if so, where a
Study Protocol	Authors stated whether or not the study protocol is available—and, if so,
Analysis Plan	Authors stated whether or not the analysis plan is available—and, if so, w
Materials Transparency	Authors stated whether or not materials are available—and, if so, where.
Data Transparency	Authors stated whether or not data are available—and, if so, where.
Analytic Code Transparency	Authors stated whether or not analytic code is available—and, if so, wher
Reporting Transparency	Authors stated whether or not they used a reporting guideline—and, if so

Table 3
Comparison of TOP Guidelines (2015 vs. 2025)

Feature	
Overall, Purpose	Focused on improving transparency in journal publications through structur
Organization of Standards	8 separate standards: Citation, Data Transparency, Analytic Code Transpa
Levels of Implementation	Level 0: Not addressed or Adopted; Level 1: Disclosure; Level 2: Requireme
Citation	Treated as its own standalone standard (citation of data, materials, and coo
Preregistration	Two standards: preregistration of studies and preregistration of analysis pla
Design & Analysis	Included as the Design and Analysis Transparency standard.
Verification	Implied verification appears only at Level 3 within each standard (e.g., inde
Replication	Replication was one of the eight standards; Registered Reports were treated
Intended Users	Designed primarily for journals.
Flexibility	Encouraged modular adoption. So, journals choose which standards and lev